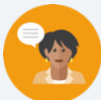


9.10 Applying Different Methods of Solving a Quadratic

Lesson Introduction

 Benchmarks	MA.912.AR.3.1 Given a mathematical or real-world context, describe, write, and solve one-variable quadratics equations over the real number system.		 Learning Targets	- I can solve a quadratic using different methods. - I can choose a method for solving a quadratic based on the context.	
 Benchmark Coherence	Building On MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.			Working Toward MA.912.AR.3.2 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real and complex number systems.	
 Essential Question	How can I know which method is most appropriate for solving a quadratic equation in a given situation?				
 Lesson Narrative	In this lesson, students will use all the methods they have learned in the previous lessons to solve a quadratic equation. Students will be given various contexts and be asked to solve a quadratic using the method that is most appropriate for the given context.				
 Component Summary	9.10.1 Warm-Up (~5 min.)	Real-world free fall exploration (<i>SE p. 108</i>)	9.10.3 Guided Instruction (10 min.)	Choosing a method for solving a quadratic equation (<i>SE p. 110</i>)	
	9.10.2 Collaboration (15 min.)	Review and use the different methods of solving quadratics (<i>SE p. 109</i>)	9.10.4 Guided Practice (10 min.)	Practice choosing a method for solving quadratic equations (<i>SE p. 112</i>)	
	9.10.5 Wrap-Up (~5 min.)	Ticket Out the Door on choosing a method for solving quadratic equations (<i>SE p. 113</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Roots - Zeros <u>Additional Terms:</u> - Quadratic formula - Factored form - Standard form - Vertex form - Zero product property		N/A		As you work through the questions prior to teaching, consider barriers for your students. This will help with determining grouping and/or pairings based on their individual needs. Consider which students would benefit from small-group instruction or other support.

 Teacher Tips	Common Misconceptions - Students may choose a method that does not work for a given situation. - Students may have calculation errors. - Students may forget how to use a chosen method.	Things to Consider Create a foldable for each method so you can have students capture what they've learned and come up with examples of when to use each one.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder The 9.10.1 Warm-Up provides students with a graph about the concept of a free fall. Consider applying “Notice and Wonder” routines on the graph, positioning students to discover the relationship between the time and speed in terms of the context.	MA.K12.MTR.3.1: Mathematical Fluency Students will be working with various contexts that represent quadratic equations. They must choose the most appropriate method for solving a quadratic equation based on the given context.
 Differentiate Instruction	9.10.2 Collaboration provides students with a collaborative experience for different methods of solving quadratics. It provides questions that assess each previously covered method. Student performance on these questions can pinpoint potential differentiation scenarios. In 9.10.2 Collaboration question 6, consider having students do a read-aloud of the content prior to filling in the blanks. This could assist auditory learners with better understanding the content. The 9.10.5 Wrap-Up provides students with a Ticket Out the Door that allows students to show what they know about choosing and using an appropriate method to solve a quadratic equation.	
Lesson Notes:		

9.10.1 Warm-Up: Free Fall

Component Overview: Students will explore a real-world example of free fall. This activity is designed to be done independently.

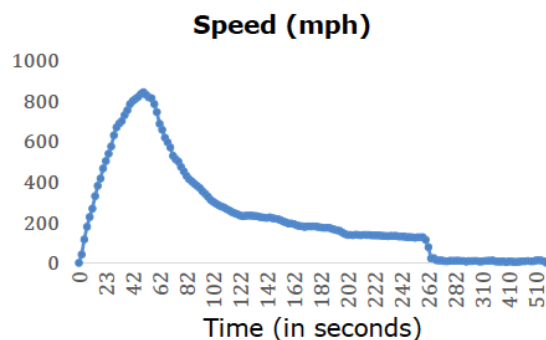


9.10.1 Warm-Up: Free Fall

Free fall is a special type of motion in which the only force acting upon an object is gravity. On October 14, 2012, Felix Baumgartner was the first person to break the sound barrier in free fall when he completed a parachute jump from a height of 127,852 feet (ft).

1. Determine how many miles Felix was above the earth when he jumped out of the capsule. 1 mile = 5,280 ft
24.214 miles

His fastest speed during the free fall was 843.6 miles per hour (mph). Felix spent 4 minutes 22 seconds in free fall and the total jump time was 9 minutes 9 seconds. The graph shows his speed, in miles per hour, during the jump.



2. What do you think happened in Felix's fall at the highest point on the graph?

Answers will vary. Felix could have opened his parachute causing his speed to slow down.



Encourage students to leverage word problem strategies as they engage in this activity. For examples of word problem strategies, refer to the Unit Guide for this unit.



Challenge students to describe what is happening throughout the course of the graph within the context.



Have students share their answer to question 2. Poll the class to determine who agrees or disagrees to determine common trends of thinking.

9.10.2 Collaboration: Different Methods of Solving Quadratics

Component Overview: Students will work collaboratively with a partner to review and use the different methods of solving quadratics. This activity is a culmination of what students have learned throughout this unit. Provide students with a recap of what they are being asked to do in this activity and release students to work on these problems together. Students should complete questions 1-3 on their own, and in question 4 work with their partner to check their work. This process should repeat for questions 5 and 6.



9.10.2 Collaboration: Different Methods of Solving Quadratics

1. Use the quadratic formula to solve $9x^2 + 36x + 36 = 0$.

$$\begin{aligned} a &= 1 \quad b = 4 \quad c = 4 \\ x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ x &= \frac{-(4) \pm \sqrt{(4)^2 - 4(1)(4)}}{2(1)} \\ x &= \frac{-(4) \pm \sqrt{16 - 16}}{2} \\ x &= \frac{-(4) \pm \sqrt{0}}{2} \\ x &= \frac{-4}{2} = -2 \end{aligned}$$

2. Solve $9x^2 + 36x + 36 = 0$ by factoring.

$$\begin{aligned} 9(x^2 + 4x + 4) &= 0 \\ 9(x + 2)(x + 2) &= 0 \\ (x + 2) &= 0 & (x + 2) &= 0 \\ x &= -2 \end{aligned}$$

3. Solve $9(x + 2)^2 = 0$.

$$\begin{aligned} \frac{9}{9}(x + 2)^2 &= \frac{0}{9} \\ (x + 2)^2 &= 0 \\ \sqrt{(x + 2)^2} &= \pm\sqrt{0} \\ x + 2 &= 0 \\ x &= -2 \end{aligned}$$

4. Check your work with your partner and resolve any differences if necessary.

Answers will vary.



Questions 1-3 provide different opportunities for targeted differentiation.

Question 1 – Quadratic Formula

- Students who get this incorrect may not understand how to use the quadratic formula appropriately

Question 2 – Solving by Factoring

- Students who get this incorrect may struggle with factoring

Question 3 – Square Roots

- Students who get this incorrect may not understand the process of using square roots to solve



Have students share their answers to question 4 while you look for class trends and share peer feedback.



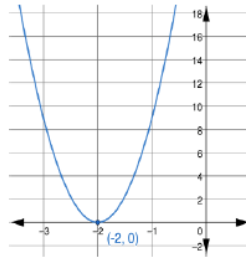
Would you have preferred to use a different method on any of the other questions? Why or why not?
Is the quadratic in question 3 the same as the quadratic seen in questions 1 and 2?

9.10.2 Collaboration: Different Methods of Solving Quadratics (continued)

5. The graph of $y = 9(x + 2)^2$ is shown.

Work with your partner to write an explanation that connects the graph of $y = 9(x + 2)^2$ to the solutions to $9x^2 + 36x + 36 = 0$ and $9(x + 2)^2 = 0$.

Answers will vary. Each equation has one solution, which is $(-2, 0)$. The graph touches the x -axis one time at $(-2, 0)$.



6. Work with your partner to complete the statements using the word bank provided.

different	factored	factored form	same
standard form	vertex form	x -intercept(s)	
y -intercepts	zero product property		

The three methods for solving quadratics use different forms of a quadratic. Each method results in the same solution(s). To solve by factoring, the quadratic is factored, then the zero product property is applied to find the solutions. To solve by completing the square, the quadratic is rewritten to vertex form. Then, the equation is solved for x . The quadratic formula is best used when the quadratic cannot be factored or if completing the square is difficult. The solution(s) are the x -intercept(s) of the parabola. These are also known as the zeros of the graph or roots of the quadratic.



The **roots** or **zeros** of a quadratic equation are the solution(s) of an equation that has the form $ax^2 + bx + c = 0$.



Connecting the visual representation of the solution continually supports student understanding of why the quadratic is equal to 0. Help students also connect this to the x -intercept of the parabola. Allow students to share their explanations. Consider polling the class to see who had similar observations and allow for peer responses to the explanations presented.



Consider having students do a read-aloud of the content prior to filling in the blanks. This could assist auditory learners with better understanding the content.



Have students do a read-aloud of this explanation. It sums up the different methods using accurate mathematical language. Consider asking students comprehension questions.



Consider putting the words with their definitions on the backs of index cards. If groups contain students with various learning styles, the cards can serve as a visual aid for the visual learner, a constant auditory stimulus for auditory learners, and a hands-on reference for kinesthetic learners.

9.10.3 Guided Instruction: Choosing a Method

Component Overview: Students will engage in guided instruction on choosing a method for solving a quadratic equation.



9.10.3 Guided Instruction: Choosing a Method

1. The equation $y = -16x^2 + 128x + 1$ models the height, in feet, after x seconds, of a toy rocket that is launched from 1 foot above the ground.

- a. What is the height of the rocket when it is on the ground?
0 feet
- b. Use the table to determine when rocket reaches its maximum height.

Time (seconds)	-2	1	3	5	7	10
Height (feet)	-319	113	241	241	113	-319

The maximum height is reached at 4 seconds.

- c. The equation $-16x^2 + 128x + 1 = 221$ can be used to determine the time the rocket was at 221 feet. Rewrite the equation by filling in the blanks.
 $-16x^2 + 128x + (-220) = 0$
- d. Consider the three methods of solving quadratics and choose one method to solve the equation.
- Completing the square
 - Factoring
 - Quadratic formula

Explain your choice.

Answers will vary based on choice of method. I chose quadratic formula because I know I can substitute in the values of a , b , and c since the equation is already set equal to 0. I chose factoring method because it looks like it can be factored easily.



For question 1 part b, model for students or remind them how to use tables to find the vertex.



Make connections between the equations and table as students continue to develop their understanding of quadratics modeling contexts using multiple representations. Consider sketching the graph as an additional representation, if helpful for your students.



For a challenge, see if students can think of another real-world scenario that would result in a similar equation or if they can interpret what the other parts of the equation represent in the context.

9.10.3 Guided Instruction: Choosing a Method (continued)

- e. Ask your partner which method they selected. Record their choice and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Method	My Summary of Their Reason	Initials
- Completing the square - Factoring - Quadratic formula	<i>Answers will vary. She chose the quadratic formula because she is good at evaluating and is comfortable setting the formula up correctly.</i>	<i>Answers will vary. TD.</i>

- f. Solve $-16x^2 + 128x + 1 = 221$ showing your work.
Method used will vary.

$$\begin{aligned}
 x &= \frac{-(128) \pm \sqrt{(128)^2 - 4(-16)(-220)}}{2(-16)} \\
 x &= \frac{-(128) \pm \sqrt{16384 - 14080}}{-32} \\
 x &= \frac{-(128) \pm \sqrt{2304}}{-32} \\
 x &= \frac{-(128) \pm 48}{-32} \\
 x &= \frac{-(128)+48}{-32} \quad x = \frac{-(128)-48}{-32} \\
 x &= 2.5 \quad x = 5.5
 \end{aligned}$$

- g. Complete the statement about the solution to $-16x^2 + 128x + 1 = 221$.

At 2.5 seconds and 5.5 seconds, the toy rocket is at a height of 221 feet.



Poll the students after completing question 1e to see how many of them would switch their methods based on their partner's rationale. Have students share what they heard from others that may have caused them to change their method.



Prepare to differentiate student support based on their method of choice as well as errors made when executing their chosen method. As students continue to develop a sense for which methods are most appropriate based on the structure of equations, consider facilitating opportunities to discuss their choices with other students. For some students, hearing the thought processes of peers will help them solidify their own understanding.

Be careful that conversations about which method(s) are appropriate do not lead students to believe there is only one correct way to solve a given equation. Any method can be used, but some equations may lend themselves better to one or more methods than another.

9.10.3 Guided Instruction: Choosing a Method (continued)

2. The function $M(x) = 5x^2 - 160x + 6179$ models the number of miles, in millions, that U.S. passengers traveled on a train for the period 1970 to 2000, where x is the number of years since 1970.

- a. Use the equation $5x^2 - 160x + 6179 = 8279$ to determine in what year U.S. passengers traveled 8,279 millions of miles.

$$x = \frac{-(-160) \pm \sqrt{(-160)^2 - 4(5)(-2100)}}{2(5)}$$

$$x = \frac{160 \pm \sqrt{25600 + 42000}}{10}$$

$$x = \frac{160 \pm 260}{10}$$

$$x = 42 \quad x = -10$$

In 2012, US passengers traveled 8,279 millions of miles.

- b. Interpret the solutions and determine if the solutions are viable.

Neither solution is viable. $x = -10$ represents 10 years before 1970, which is 1960, and $x = 42$ represents 42 years since 1970 which is 2012. Both of those years are outside of the domain of the model.



Consider using a word problem strategy for this question, prior to moving into the question. For examples of strategies, refer to the Unit Guide for this unit.



How do you use the given time period to help determine whether the solution is viable?

9.10.4 Guided Practice: Choosing a Method

Component Overview: Students will engage in guided practice on choosing a method for solving quadratic equations. Read the context for the problem in question 1 and release students to try parts a and b on their own. Call students back together to review question 1 parts a and b as a whole group.



9.10.4 Guided Practice: Choosing a Method

1. The function $f(x) = -3x^2 + 54x + 83$ models the sales of music cassette tapes for the years 1980 to 2000, in tens of millions of dollars, where x is the number of years since 1980.



- a. Use the equation $-3x^2 + 54x + 83 = 0$ to determine when the sales of music cassette tapes are \$0.

$$x \approx 19.424 \quad x \approx -1.424$$

In the year 1999, the sales of music cassette tapes were \$0.

- b. Interpret the solutions and determine if the solutions are viable.

Answers will vary. $x = -1.424$ is not viable because it would be for 1978, which is outside of the domain of the model. $x = 19.424$ means in the year 1999, the sales of music cassette tapes were \$0.



Students may remember this question from earlier in the unit. This question asks that the student determines when the sales are \$0. Students may wonder about the reality of this. Discuss with them that the technology had to be introduced at some point, and although used cassette tapes may be found today the music industry no longer sells them. Students may know that vinyl records were once off the market but have made a comeback. Ask students if they think cassette tapes will make a comeback as well.

Ask students which method they would choose to solve the question.



Have students share their answers to question 1 part b to discuss viability of answers.

9.10.5 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door that assesses their understanding of choosing a method for solving quadratic equations. This activity should be done independently so that you have data on how students understand the content. Students should be allowed to access a calculator to complete the Wrap-Up.



9.10.5 Wrap-Up: Ticket Out the Door

The function $S(x) = -0.5x^2 + 50x - 8$, where x is time in seconds (s), models the speed, S , in kilometers per hour (km/h) of Felix Baumgartner during the first 90 seconds of his record-breaking free fall.



Student performance on this Wrap-Up will provide data that can be used for differentiation or additional support.

1. Use the function, $S(x)$, to determine when Felix reached the greatest speed.

$$S(x) = -0.5(x^2 - 100x) - 8$$

$$S(x) = -0.5(x^2 - 100x + 2500) - 8 + 1250$$

$$S(x) = -0.5(x - 50)^2 - 8 + 1250$$

$$S(x) = -0.5(x - 50)^2 + 1242$$

The greatest speed was reached 50 seconds after jumping.

2. Mathematical models use data to determine an equation that closely fits but may not be exact. According to the actual data collected during Felix's fall, the greatest speed he reached was 1355 km/h, which occurred at 50 seconds. How well did the function $S(x)$ match the real data?

Answers will vary. The function $S(x)$ showed that the greatest speed at 50 seconds would be 1242 km/h. The function was 113 km/h off.